End-to-End Block-Sparse Featurizers for a Frozen Vision Readout

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Engineering smoke experiment: one seed, small data and dictionary; no confirmatory claim.

Abstract

We study whether output-distribution matching can train a block-sparse representation that preserves a fixed vision classifier. A Grassmannian block-sparse featurizer replaces final DINOv3 patch activations. Its parameters are initialized randomly and optimized using forward KL divergence from the original composite classifier to the reconstructed classifier. Local reconstruction, scalar sparsity and a mixed objective provide comparisons. Mean paired local-minus-KL difference: 0.115604 nats/image over 1 seed(s). KL training had lower mean output divergence. A seed-level confidence interval cannot be estimated from one seed. These measurements concern the frozen ten-class readout; they do not establish semantic concept recovery.

1. Context, motivation and related work

Local sparse dictionary learning rewards activation reconstruction, whereas end-to-end sparse dictionary learning [1] rewards preservation of downstream output distributions. Block-sparse featurizers [2] group signed coordinates into low-dimensional charts, motivated by superposed manifolds. We combine the released Grassmannian implementation with the executable forward-KL convention of [1]. Prior work studies short KL fine-tuning [3]; this experiment instead trains every BSF condition from scratch. Later BSF analysis [4] motivates caution about splitting and stability.

DINOv3 [5] produces embeddings. Consequently, the original model here is explicitly DINOv3 plus one independently fitted, then frozen, patch-only Imagenette classifier. The dataset and classifier are smaller adaptations of [2], not a reproduction of its complete ImageNet evaluation. The model was pretrained externally; held-out claims refer to the new featurizer and classifier procedure.

2. Method

Let H contain the final 196 patch vectors of width 768, excluding CLS and four register tokens. Fit a positional mean mu and scalar RMS s on training images. For x=(H-mu)/s, QR produces orthonormal frames D_g. The tied encoder computes z_g=gamma_g x D_g^T, retains the k blocks with largest Euclidean norms, and reconstructs x_hat=sum_g z_g D_g. Insert H_hat=s x_hat+mu without adding the reconstruction residual. The released code uses an unconstrained learned gain per block and differentiable QR every forward; these details differ from the paper's shared positive gain and periodic projection description. We reuse a single QR result within each forward.

Pool each activation coordinate by its maximum absolute magnitude over patches. Fit a ten-class multinomial logistic readout on clean training activations, choosing C in {0.01,0.1,1,10} by development cross-entropy. Freeze the head for all conditions. With p=softmax(head(H)) and q=softmax(head(H_hat)), L_KL=mean_images sum_classes p(log p-log q). Teacher targets are detached, temperature is one, and log-softmax/KL use float32. Only the BSF is optimized.

Main dictionary: 4096 blocks of width 3, keeping 16 blocks. The scalar ablation uses 12288 width-one blocks and 48 active coordinates. The smaller smoke configuration, if used in this report, is explicitly listed below. Scalar controls match total/active coordinates but have more gain parameters. They are tied signed featurizers, not

ReLU SAEs. A mixed objective adds 0.1 times normalized patch MSE; the primary KL condition has no reconstruction penalty.

3. Data, optimization and analysis

Image SHA-256 and 64-bit perceptual hashes (Hamming distance at most four) define duplicate components before splitting. Official Imagenette training becomes an 80/20 grouped training/development split at seed 42; official validation is reserved for testing. Training members of test-related duplicate groups are excluded. Test groups inspected during the separate engineering smoke run are also excluded from the full study's test set. Normalization never refits on development or test. Perceptual hashing is a limited screen, not a guarantee that all related images are detected.

Adam uses gradient clipping at one and cosine decay to 10% of the initial learning rate. The pilot compares 1e-4 and 3e-4 equally, using independent seeds 0,1,2 on 510 training and 250 development images. A paired-difference power calculation targets 80% power for a 10% relative KL reduction, with 5-10 confirmatory seeds and at most 20 epochs. The measured time forecast includes a 25% overhead allowance. Every condition gets the same confirmatory image exposure. The 24-hour bound cannot be extended merely to obtain significance.

Primary contrast is per-seed mean local-MSE KL minus KL-trained KL, paired by initialization and data order. Positive differences favor output training. We report Student 95% intervals across seeds and separate image-group bootstrap intervals. Correlated patches are not independent replicates. Small pilot variance samples limit power estimates. Identity, mean replacement and untrained BSF controls test the bottleneck and establish reference outcomes.

Recorded configuration: {'profile': 'smoke', 'groups': 32, 'block_size': 3, 'active': 4, 'epochs': 3, 'microbatch': 2, 'effective_batch': 8}

4. Results and claims

Mean paired local-minus-KL difference: 0.115604 nats/image over 1 seed(s). KL training had lower mean output divergence. A seed-level confidence interval cannot be estimated from one seed.

Condition	Seeds	KL nats/image	Accuracy	NMSE	Dead blocks
Block KL + 0.1 MSE	1	5.75317	14.00%	0.96633	0.00%
Block KL	1	5.75244	14.00%	0.96635	0.00%
Block MSE	1	5.86805	10.00%	0.96103	0.00%
Scalar KL	1	5.13261	26.00%	0.94416	0.00%
Scalar MSE	1	5.3282	26.00%	0.937	0.00%

Control identity: KL 0 nats/image, accuracy 100.00%, NMSE 0.

Control mean: KL 6.98686 nats/image, accuracy 10.00%, NMSE 1.00872.

Control untrained: KL 6.02086 nats/image, accuracy 10.00%, NMSE 0.969882.

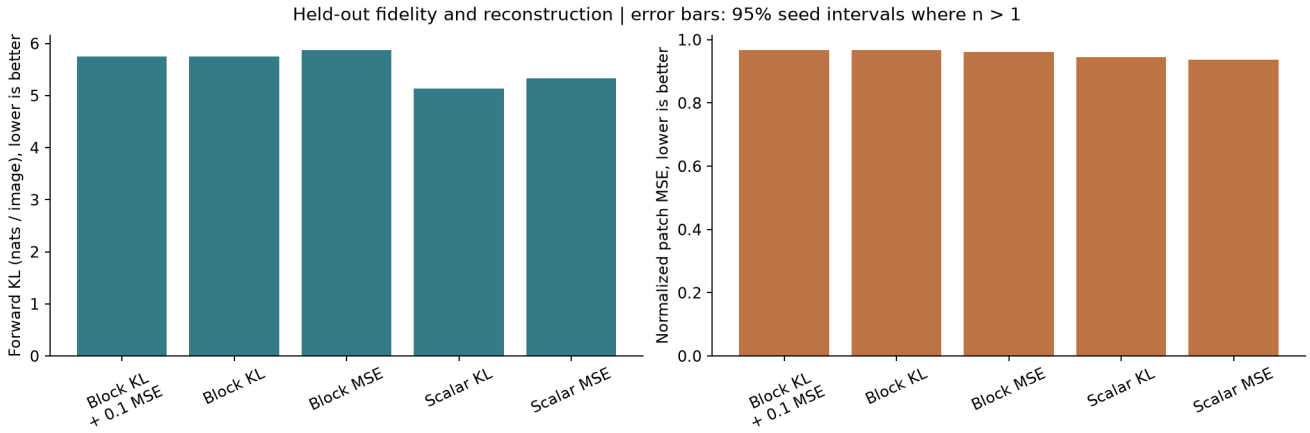


Figure 1. Measured held-out output fidelity and patch reconstruction. Seed error bars appear only when more than one seed is available; their absence is not evidence of certainty.

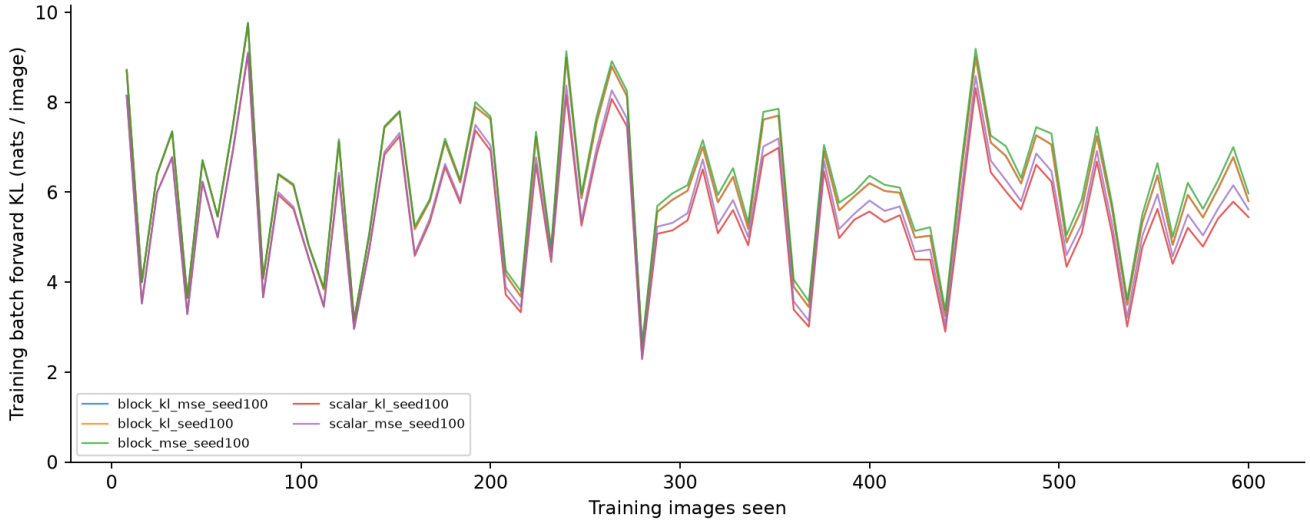


Figure 2. All displayed completed runs' training-batch KL against images seen. Batch fluctuations are retained; no post-hoc smoothing or favorable epoch selection is used.

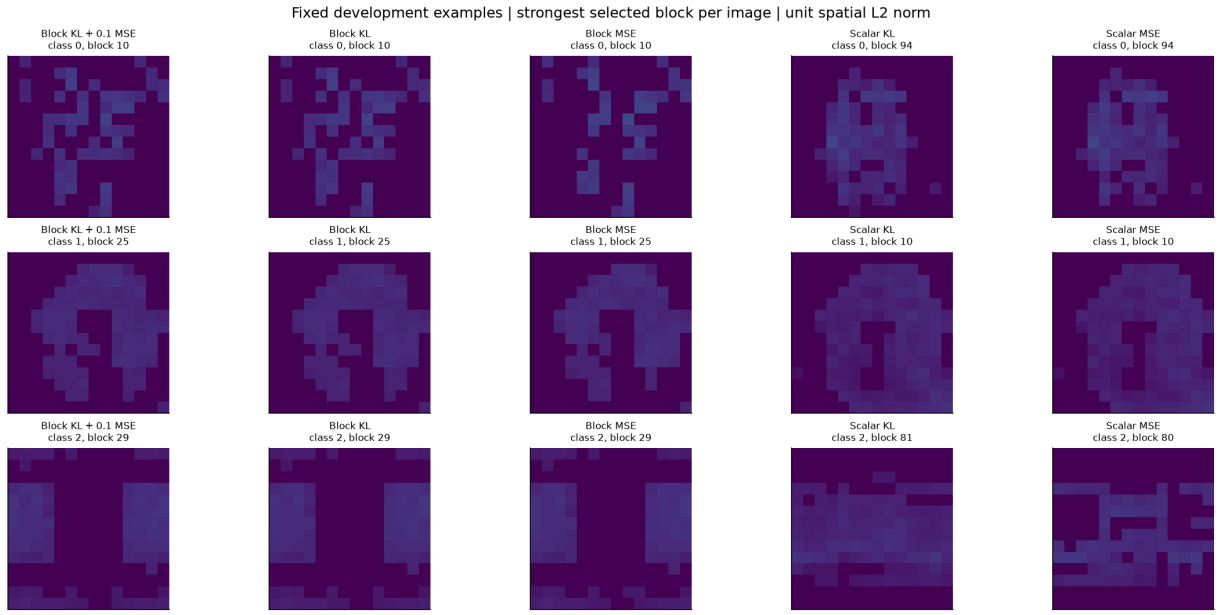


Figure 3. Algorithmically selected development feature maps for fixed examples. Maps are normalized separately to unit spatial L2 norm. Feature indices differ across independently fitted dictionaries; no semantic correspondence is assumed.

5. Interventions and geometry

Eight blocks are ranked by development energy. For fixed test examples, remove the strongest selected block, remove a random active block among the eight closest-energy alternatives, or replace that block's complete patch code with a predetermined same-class/other-class donor. These are causal interventions on the composite classifier. Changes in KL measure behavior under these interventions and do not show that the block encodes the class or another semantic factor. Complete per-example energies, donors and outputs accompany the report.

block_kl_mse, seed 100, 20 examples: remove_selected: 0.14571; remove_random_matched: 0.060455; swap_same_class: 0.039217; swap_other_class: 0.095673 (mean KL change).

block_kl, seed 100, 20 examples: remove_selected: 0.14686; remove_random_matched: 0.060519; swap_same_class: 0.038722; swap_other_class: 0.095948 (mean KL change).

block_mse, seed 100, 20 examples: remove_selected: 0.11457; remove_random_matched: 0.052537; swap_same_class: 0.024701; swap_other_class: 0.080206 (mean KL change).

scalar_kl, seed 100, 20 examples: remove_selected: 0.093861; remove_random_matched: 0.038035; swap_same_class: 0.049652; swap_other_class: 0.065602 (mean KL change).

scalar_mse, seed 100, 20 examples: remove_selected: 0.045187; remove_random_matched: 0.038714; swap_same_class: 0.0053338; swap_other_class: 0.022665 (mean KL change).

6. Discussion, limitations and implications

A ten-class softmax constrains at most nine independent logit contrasts per image. Output preservation can therefore coexist with poor reconstruction, altered spatial maps or discarded information. This study does not test broad DINO transfer, segmentation, depth or diffusion behavior. The fixed probe and pooling define the measured function. Max pooling sends gradients only through winning patch coordinates; hard TopK supplies no ordinary gradient to unselected blocks for a given example. Dead blocks and seed sensitivity are outcomes, not hidden implementation successes.

Float32 cached final patches make BSF gradients equivalent to online frozen-backbone extraction within measured numerical tolerance. This does not exercise backpropagation through earlier DINO layers. Dense

kernels remain in use, so this work makes no speedup claim. Pilot power estimates may be optimistic or pessimistic for longer runs. Any budget failure, missing seed or post-hoc correction must be reported with the saved status and provenance.

7. Conclusion

The implementation supplies a reproducible test of output-KL training for structured sparse vision codes. Mean paired local-minus-KL difference: 0.115604 nats/image over 1 seed(s). KL training had lower mean output divergence. A seed-level confidence interval cannot be estimated from one seed. Claims are restricted to the measured teacher, data, architecture and compute allocation.

References

1. Braun et al. Identifying Functionally Important Features with End-to-End Sparse Dictionary Learning (2024)
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6. Imagenette dataset source
7. Goodfire implementation, revision 219f121
8. Apollo implementation, revision 83b4add
9. SAEBench: A Comprehensive Benchmark for Sparse Autoencoders (2025)
10. Card et al. With Little Power Comes Great Responsibility (2020)
11. Lones. How to avoid machine learning pitfalls (2021)

Appendix: reproducibility and AI statement

Code, preregistration, lockfile, run tables and model releases: project repository. Per-run checkpoints bind preprocessing, the frozen classifier, source revisions and data manifests. Original images and gated backbone weights are not redistributed.

AI statement: Codex wrote this codebase and paper. Human cleanup and rewriting are in progress. A human provided the exact specification that Codex executed.